Integer linear programming

Outline

Why integer linear programming?

Canonical maximization form
- Feasible region and objective function
- Optimal solution and optimal value
- Graphical solution method

Possible outcomes

Complete enumeration

Typical mistakes

In summary

Learning objectives

At the end of this chapter you will be able to:

- write an ILP problem in canonical maximization form, and distinguish ILP, BIP and MIP problems;
- draw the feasible region of a small ILP problem and find an optimal solution graphically;
- explain why the feasible region is not convex, and what the three outcomes of an ILP problem are;
- explain why complete enumeration fails, and how much it costs;
- say why bounds on the optimal value are needed to avoid enumeration.

Prerequisites

From the module *Linear Programming*:

- Chapter 2 «Half-spaces and polyhedra»
- Chapter 4 «Linear Programming problems»: the canonical maximization form, the feasible region, optimal solutions and the optimal value
- Chapter 5 «Possible outcomes and equivalent formulations»: infeasible, unbounded and optimal problems

Why integer linear programming?

Introduction: Why integer linear programming?

- Many real decisions cannot be split: how many cars to build, whether to open a plant or not
- We see what an integer linear programming problem is, and why it is hard
- We meet the car example, the guiding example of the whole module, and a first challenge

Discrete decisions

- Many decisions are **discrete**: a quantity that cannot be split, a yes/no choice
- The number of possible solutions is finite, but it can be huge
- An **integer linear programming** problem describes such a decision with linear constraints and a linear objective function, as in linear programming, but some variables must take integer values
- A variable restricted to $\{0, 1\}$ is a yes/no decision (*in* or *out*)

The question of the module

- The integrality requirement makes the problem much harder than its continuous version
- The whole module answers one question:

How do we find an optimal solution,
and how do we certify that it is optimal?

Certifying optimality

- A solution is certified optimal when a **lower bound** (a feasible solution) and an **upper bound** (a proof that nothing is better) have the same value
- Every method of this module produces one of the two bounds, or both

A first model: the car example

- The variables x_1 and x_2 are the numbers of cars of two types: they are non-negative integers, and from now on the problem is a mathematical object
- The ILP problem reads:

$$
\begin{array}{rrcrcr}
\max & 3\,x_1 & + & 2\,x_2 & & \\
\text{subject to} & 10\,x_1 & + & 4\,x_2 & \leq & 45 \\
& 4\,x_1 & + & 14\,x_2 & \leq & 49 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$

A challenge

Before going on, find a **good feasible solution**
of the car example.

Then try to answer the harder question:
how can you **prove** that no better solution exists?

- The car example is the guiding example of the whole module, and each chapter will give a piece of the answer

Summary

- Discrete decisions lead to linear models with **integer variables**
- Integrality makes the problem much harder than its continuous version
- Optimality is certified by a **lower bound** and an **upper bound** with the same value
- The **car example** will follow us through the whole module

Canonical maximization form

Introduction: Canonical maximization form

- We define an ILP problem in canonical maximization form, as for linear programming
- We define its feasible region, its optimal solutions and its optimal value
- We solve the car example graphically: a first, small optimal solution
- These definitions are the language of the whole module

Definition 1: Integer Linear Programming problems in canonical maximization form (1/3)

Given:

1. *an objective-function vector* $\boldsymbol{c} \in \mathbb{Z}^n$
2. *a constraint matrix* $\boldsymbol{A} \in \mathbb{Z}^{m \times n}$
3. *a right-hand-side (RHS) vector* $\boldsymbol{b} \in \mathbb{Z}^m$ *with* m *rows*

$$\underbrace{\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}}_{\boldsymbol{c}} \in \mathbb{Z}^n \quad \underbrace{\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}}_{\boldsymbol{A}} \in \mathbb{Z}^{m \times n} \quad \underbrace{\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}}_{\boldsymbol{b}} \in \mathbb{Z}^m$$

Definition: Integer Linear Programming problems in canonical maximization form (2/3)

*an **Integer Linear Programming problem** ILP in **canonical maximization form** with n non-negative integer variables*

$$\underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \in \mathbb{Z}^n_{\geq 0}$$

and m less-than-or-equal-to linear constraints is the optimization problem of the next slide.

Definition: Integer Linear Programming problems in canonical maximization form (3/3)

The ILP problem in canonical maximization form is the following optimization problem:

$$\max \underbrace{\sum_{j=1}^{n} c_j x_j}_{\boldsymbol{c'x}}$$

$$\textit{subject to} \quad \underbrace{\sum_{j=1}^{n} a_{ij} x_j \leq b_i, \qquad \forall i \in \{1, 2, \ldots, m\},}_{\boldsymbol{Ax \leq b}}$$

$$\underbrace{x_j \in \mathbb{Z}_{\geq 0}, \qquad \forall j \in \{1, 2, \ldots, n\}}_{\boldsymbol{x} \in \mathbb{Z}^n_{\geq 0}} .$$

Notation

- The data $\boldsymbol{c}$, $\boldsymbol{A}$ and $\boldsymbol{b}$ are **integer**, as in the whole module
- With a slight abuse of notation, we write the ILP problem in compact form as

$$\max \left\{ \boldsymbol{c}'\boldsymbol{x} \ : \ \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}, \ \boldsymbol{x} \in \mathbb{Z}^n_{\geq 0} \right\}$$

- The set $\mathbb{Z}^n$ of the points of $\mathbb{R}^n$ with integer components is called the **integer lattice**

An LP problem with integer variables

An ILP problem is an LP problem
in canonical maximization form
with the additional requirement $\boldsymbol{x} \in \mathbb{Z}^n$:

the linear function $\boldsymbol{c}'\boldsymbol{x}$ is maximized
over the points of the **integer lattice**
that satisfy the constraints.

BIP and MIP problems

- If all the variables are restricted to $\{0, 1\}$, the problem is a **binary integer linear programming** (BIP) problem
- If only some of the variables are integer, the problem is a **mixed integer linear programming** (MIP) problem
- Constraints $\geq$ and $=$, and minimization problems, are brought to the canonical maximization form as in linear programming

Example (1/2)

- Consider the following input data of an ILP problem with $n = 2$ and $m = 2$:

$$\boldsymbol{c} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \qquad \boldsymbol{A} = \begin{pmatrix} 10 & 4 \\ 4 & 14 \end{pmatrix} \qquad \boldsymbol{b} = \begin{pmatrix} 45 \\ 49 \end{pmatrix}$$

- The numerical entries of the two column vectors and the matrix are

$$c_1 = 3, \; c_2 = 2,$$
$$a_{11} = 10, \; a_{12} = 4, \; a_{21} = 4, \; a_{22} = 14,$$
$$b_1 = 45, \; b_2 = 49.$$

Example (2/2)

- The associated ILP problem in canonical maximization form, with $n = 2$ non-negative integer variables and $m = 2$ less-than-or-equal-to linear constraints, reads:

$$\begin{array}{rrcrcr} \max & 3\,x_1 & + & 2\,x_2 & & \\ \text{subject to} & 10\,x_1 & + & 4\,x_2 & \leq & 45 \\ & 4\,x_1 & + & 14\,x_2 & \leq & 49 \\ & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0} \end{array}$$

- The variables x_1 and x_2 can be read as the numbers of cars of two types to produce: this ILP problem is the **car example**, the guiding example of the module

Now that we have the problem, we describe its two ingredients: the set of solutions among which we choose (**feasible region**) and the function that we maximize (**objective function**).

Definition 2: feasible region and objective function (1/2)

*The **feasible region** of an ILP problem is the set F defined as follows:*

$$F = \left\{ \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \in \mathbb{Z}^n : \underbrace{\sum_{j=1}^{n} a_{ij}\, x_j \leq b_i, \quad \forall i \in \{1, 2, \ldots, m\}}_{\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}}, \right.$$

$$\left. \underbrace{x_j \geq 0, \quad \forall j \in \{1, 2, \ldots, n\}}_{\boldsymbol{x} \geq 0} \right\}.$$

Definition: feasible region and objective function (2/2)

*The **objective function** of an ILP problem, to be maximized, is the (scalar) **linear function** f defined as follows:*

$$f : \mathbb{Z}^n \to \mathbb{R}, \qquad f : \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\boldsymbol{x}} \mapsto \underbrace{\sum_{j=1}^{n} c_j x_j}_{\boldsymbol{c'x}}$$

The integer points of a polyhedron

- Let
$$P = \{ \boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b},\ \boldsymbol{x} \geq 0 \}$$
be the polyhedron of the constraints, that is, the feasible region of the LP problem with the same data

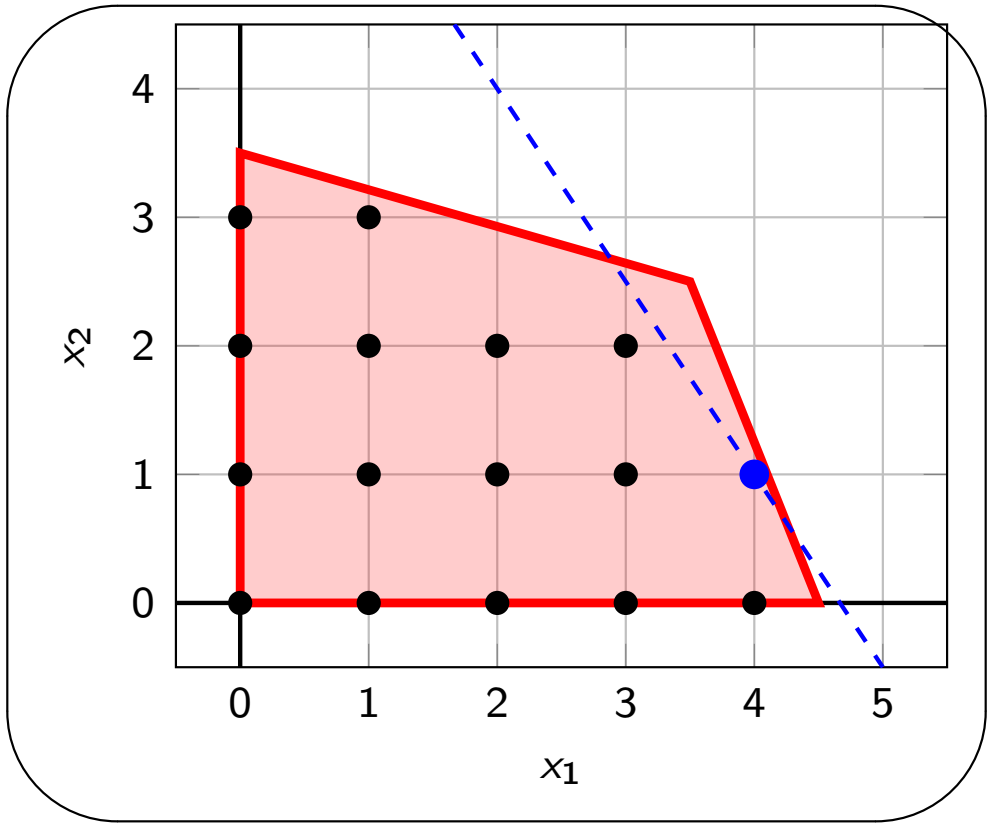
- The feasible region of the ILP problem is
$$F = P \cap \mathbb{Z}^n$$

- The set F is made of isolated points of the lattice, hence it is **not convex**

Example (1/3)

- Consider the ILP problem of the car example
- The feasible region F is the set of the 16 black points of the figure on the next slide
- They are the points of the lattice $\mathbb{Z}^2$ in the polyhedron P (red)

Example (2/3)



The feasible region is the set of the 16 integer points (black) inside the polyhedron (red). The optimal solution $(4, 1)$ is blue, with the objective line $3\,x_1 + 2\,x_2 = 14$ (dashed).

Example (3/3)

- The point $\bar{\boldsymbol{x}} = (2, 2)$ belongs to F, because

$$10 \cdot 2 + 4 \cdot 2 = 28 \leq 45 \qquad \text{and} \qquad 4 \cdot 2 + 14 \cdot 2 = 36 \leq 49$$

- The point $(4, 2)$ does not belong to F, because

$$10 \cdot 4 + 4 \cdot 2 = 48 > 45$$

- The point $(\frac{7}{2}, \frac{5}{2})$ belongs to P but not to F, because it is not integer

Summary: feasible region

- The feasible region of an ILP problem is $F = P \cap \mathbb{Z}^n$: the integer points of the polyhedron P of the LP problem with the same data
- F is made of isolated points: it is **not convex**
- In the car example, F has 16 points

Among the points of F we look for the best one:
we now define what an **optimal solution** is.

Optimal solution and optimal value

Definition 3: optimal solution and optimal value

- *A point* $\tilde{\boldsymbol{x}}(\mathrm{ILP}) \in F$ *is an* ***optimal solution*** *of the ILP problem if*

$$\tilde{\boldsymbol{x}}(\mathrm{ILP}) \in \arg\max \left\{ \boldsymbol{c}'\boldsymbol{x} \ : \ \boldsymbol{x} \in F \right\}.$$

- *The value*

$$z(\mathrm{ILP}) = \underbrace{\max \left\{ \boldsymbol{c}'\boldsymbol{x} \ : \ \boldsymbol{x} \in F \right\}}_{=\boldsymbol{c}'\tilde{\boldsymbol{x}}(\mathrm{ILP})}$$

is the ***optimal value*** *of the ILP problem.*

Feasible and optimal solutions

- A point $\bar{\boldsymbol{x}} \in F$ is a **feasible solution** of the ILP problem
- An optimal solution is a feasible solution with the largest value of the objective function

The goal of an ILP problem is to **find an optimal solution** $\tilde{\boldsymbol{x}}(\mathrm{ILP})$, whenever it exists,
rather than merely determining the optimal value $z(\mathrm{ILP})$.

Graphical solution method

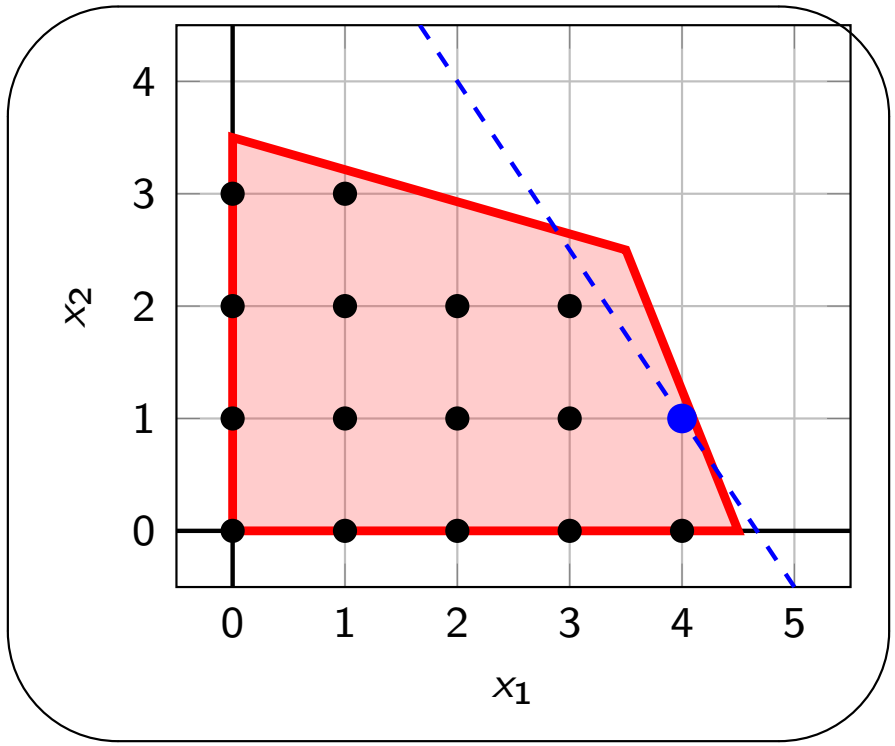
- With two variables we can draw everything, and find an optimal solution by eye

An optimal solution of an ILP problem with two variables can be graphically determined by moving the level line of the objective function in the direction of its gradient, until the last point of the lattice in the polyhedron P is reached.

Example (1/2)

- Consider the ILP problem of the car example
- The gradient of the objective function is $\nabla f = (3, 2)$
- Moving the level line $3\,x_1 + 2\,x_2 = z$ in the direction of ∇f, the last point of F reached is $(4, 1)$, on the dashed line $3\,x_1 + 2\,x_2 = 14$ of the figure of the previous example
- No other point of F lies on this line

Example (2/2)



- The car example has the **unique** optimal solution $\tilde{\boldsymbol{x}}(\mathrm{ILP}) = (4, 1)$, with optimal value $z(\mathrm{ILP}) = 14$

Summary

ILP problems:

- $\max\{\, \boldsymbol{c}'\boldsymbol{x} : \boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b},\ \boldsymbol{x} \in \mathbb{Z}^n_{\geq 0} \,\}$, with integer data: an LP problem plus integrality
- BIP: all variables in $\{0, 1\}$; MIP: only some variables integer

Feasible region and optimum:

- $F = P \cap \mathbb{Z}^n$, isolated points, not convex
- The goal is an optimal solution $\tilde{\boldsymbol{x}}(\mathrm{ILP})$, not only the value $z(\mathrm{ILP})$
- Car example: unique optimal solution $(4, 1)$, with $z(\mathrm{ILP}) = 14$

Possible outcomes

Introduction: Possible outcomes

- Not every ILP problem has an optimal solution
- We see the three possible outcomes, as in linear programming
- Two small examples show that integrality can change the outcome

Three outcomes

- As in linear programming, an ILP problem has exactly one of three **outcomes**:
  - it is **infeasible** ($F = \emptyset$)
  - it is **unbounded** (the objective function is not bounded from above on F)
  - it has an **optimal solution**

- If the polyhedron P is bounded, then F is a **finite set**, and an optimal solution exists as soon as $F \neq \emptyset$

Example: an infeasible ILP problem with a feasible LP problem

- The ILP problem

$$\max \left\{ x_1 \ : \ 2\,x_1 \leq 1, \ \ -2\,x_1 \leq -1, \ x_1 \in \mathbb{Z}_{\geq 0} \right\}$$

  is infeasible, because the constraints require $x_1 = \frac{1}{2}$

- The LP problem with the same data is feasible, with the unique feasible solution $x_1 = \frac{1}{2}$

Example: an unbounded ILP problem

- The ILP problem

$$\max \left\{ x_1 + x_2 \ : \ x_1 - x_2 \leq 1, \ x_1, x_2 \in \mathbb{Z}_{\geq 0} \right\}$$

  is unbounded

- The points (k, k), $k \in \mathbb{Z}_{\geq 0}$, are feasible, with value $2k$

Summary

- An ILP problem is **infeasible**, **unbounded** or has an **optimal solution**
- If P is bounded, F is finite and an optimal solution exists as soon as $F \neq \emptyset$
- A feasible LP problem can have an **infeasible** ILP version

Complete enumeration

Introduction: Complete enumeration

- When F is finite, the simplest method is to try every feasible solution
- On the car example it works; we then measure how much it costs in general
- The numbers show why we need something better: bounds

Enumerating the car example

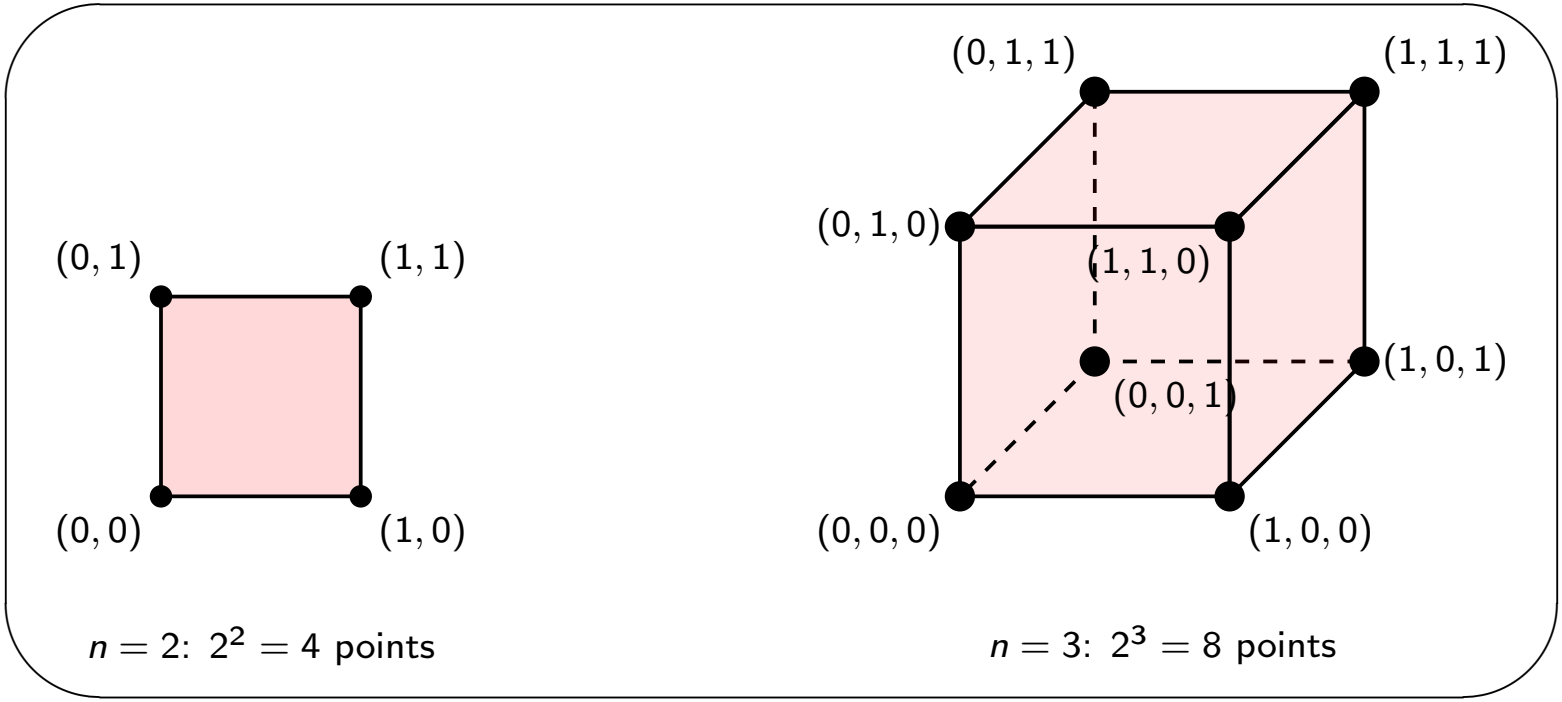
- When the feasible region is a finite set we can find an optimal solution by **complete enumeration**: we compute the objective function for every feasible solution and keep the best
- For the car example, there are 16 points:

x_1	x_2	z	x_1	x_2	z
0	0	0	3	1	11
1	0	3	**4**	**1**	**14**
2	0	6	0	2	4
3	0	9	1	2	7
4	0	12	2	2	10
0	1	2	3	2	13
1	1	5	0	3	6
2	1	8	1	3	9

The method works on tiny models

- The best value is 14, in $(4, 1)$ (in bold in the table)
- Here the method works, because the model is tiny
- Consider now the simplest case: n **binary variables** and no other constraint
- Every variable takes 2 values, so there are 2^n solutions: the vertices of an n-dimensional hypercube

The 2^n vertices of a hypercube



The solutions of n binary variables without other constraints: the 2^n vertices of an n-dimensional hypercube, for $n = 2$ (a square) and $n = 3$ (a cube).

The number 2^n explodes

- With $n = 80$ variables there are about $1.2 \cdot 10^{24}$ solutions, comparable to the estimated number of stars in the universe
- With $n = 266$ variables there are about $1.2 \cdot 10^{80}$ solutions, comparable to the estimated number of atoms in the observable universe
- Suppose that a computer examines 10^9 solutions per second, and then a computer 1000 times faster, 10^{12} solutions per second: how long does it take to enumerate 2^n solutions?

The time needed to enumerate 2^n solutions

n	2^n	at 10^9 solutions/s	at 10^{12} solutions/s
20	$\approx 1.05 \cdot 10^6$	$\approx 1.05\,\mathrm{ms}$	$\approx 1.05\,\mu\mathrm{s}$
30	$\approx 1.07 \cdot 10^9$	$\approx 1.07\,\mathrm{s}$	$\approx 1.07\,\mathrm{ms}$
40	$\approx 1.10 \cdot 10^{12}$	$\approx 18.3\,\mathrm{min}$	$\approx 1.10\,\mathrm{s}$
50	$\approx 1.13 \cdot 10^{15}$	$\approx 13.0\,\mathrm{days}$	$\approx 18.8\,\mathrm{min}$
60	$\approx 1.15 \cdot 10^{18}$	$\approx 36.5\,\mathrm{years}$	$\approx 13.3\,\mathrm{days}$
70	$\approx 1.18 \cdot 10^{21}$	$\approx 374\,\mathrm{centuries}$	$\approx 37.4\,\mathrm{years}$
80	$\approx 1.21 \cdot 10^{24}$	$\approx 3.8 \cdot 10^5\,\mathrm{centuries}$	$\approx 383\,\mathrm{centuries}$

Enumeration is impractical

Complete enumeration is impractical
already for models of moderate size.

A computer 1000 times faster
gains only about 10 variables,
because $2^{10} = 1024 \approx 1000$.

The way out: avoid enumerating

- The way out is to **avoid enumerating**
- If we know a lower bound and an upper bound on the optimal value, we can exclude entire regions of feasible solutions without visiting them
- The two bounds come from two different sources, the **feasible solutions** and the **relaxations**
- They are the subject of the next chapter, «Primal and dual bounds»

Summary

- Complete enumeration evaluates every feasible solution: on the car example it finds $(4, 1)$ among 16 points
- With n binary variables there are 2^n solutions: with $n = 80$ a computer at 10^{12} solutions per second needs about 383 centuries
- A faster computer gains only a few variables
- Bounds on the optimal value let us exclude regions without visiting them

Typical mistakes

Introduction: Typical mistakes

- We list the mistakes that are made most often with ILP problems
- They concern the shape of the feasible region, its size, the link with the LP problem and optimality
- Knowing them helps to read and solve the exercises correctly

Typical mistakes

Observation 1: Typical mistakes (1/2)

- ***Believing that the feasible region is convex.*** *It is the set of the integer points of a polyhedron, not the polyhedron. The midpoint of two feasible solutions can be infeasible.*
- ***Believing that the feasible region is always finite.*** *It is finite only if the polyhedron is bounded; otherwise the ILP problem can be unbounded.*

Typical mistakes

Observation: Typical mistakes (2/2)

- ***Believing that a feasible LP problem gives a feasible ILP problem.*** *The LP problem can have solutions while no integer point satisfies the constraints.*
- ***Taking a good feasible solution as optimal.*** *Without a bound from the other side (next chapter) there is no certificate of optimality.*

Summary

Typical mistakes:

- F is not convex, and it is finite only if P is bounded
- A feasible LP problem does not guarantee a feasible ILP problem
- A good feasible solution is not a certificate of optimality

In summary

Introduction: In summary

- We collect the key ideas of the chapter
- We close with the dashboard of the car example: the bounds found so far

Summary (1/2)

- An ILP problem maximizes a linear function over the integer points of a polyhedron; its feasible region is not convex, and it is finite when the polyhedron is bounded
- The outcomes are infeasible, unbounded or optimal, and a feasible LP problem can have an infeasible ILP version
- Complete enumeration costs 2^n evaluations for n binary variables: it is not an option beyond very small models

Summary (2/2)

- To avoid enumeration we need bounds on $z(\mathrm{ILP})$ from both sides: feasible solutions and relaxations, the subject of the next chapter

Dashboard of the car example.

- No bounds yet: the next chapter gives the first lower and upper bounds